Cable entry. F1 protects the main board's 5 V. R42, R43 not fitted.

Coin Count

D16–D19 B5819W: coin inputs
up to 40 V

J3: the phone's coin cable
J12: screw terminal + coin 4

Stage 1 controller

IO expander

Debug LEDs (green, dim, near VIO >= 3 V):
D8–D10 on while a coin switch is closed.
D11–D13 while EN_HV / POL / FIRE are high.
VIO = host I2C logic level (3.0–5.5 V).
JP1: bridge only for a 5 V–only host.

Coin 4 (U1 P7)

Stage 2 controller

J14 HV boost: no shunt = U3 unpowered, HV about 19 V;
shunt fitted = the J7–J9 voltage. R53 holds Q6 off
while U3 has no power

Polarity relay

Needs 10V on gate

HV indicator: D7 lights above about 24 V (D5 22 V + LED).
It runs from C1 itself, so it stays lit
until C1 has drained below 24 V, even
with the board unpowered.

HV SET (fit jumpers)
no J14 19 V
J14 52 V
J14 J9 60 V
J14 J8 78 V
J14 J8 J9 85 V
J14 J7 87 V
J14 J7 J9 94 V
J14 J7 J8 112 V
J14 J7 J8 J9 119 V

Voltage Measurement

J13: spare analog in (AIN3)
0 to VIO, survives +–24 V

J10 on: J11 pin 2 = HV (2–coil relay).
J2 pin 2 (red) is not connected

H1 MountingHole_M3
H2 MountingHole_M3
H3 MountingHole_M3
H4 MountingHole_M3

Sheet: /
File: Payphone_coin_board_v1.kicad_sch
Title:
Size: A3 Date: Rev:
KiCad E.D.A. 10.0.3 Id: 1/1